Learning Outcomes

In this chapter, you'll learn…

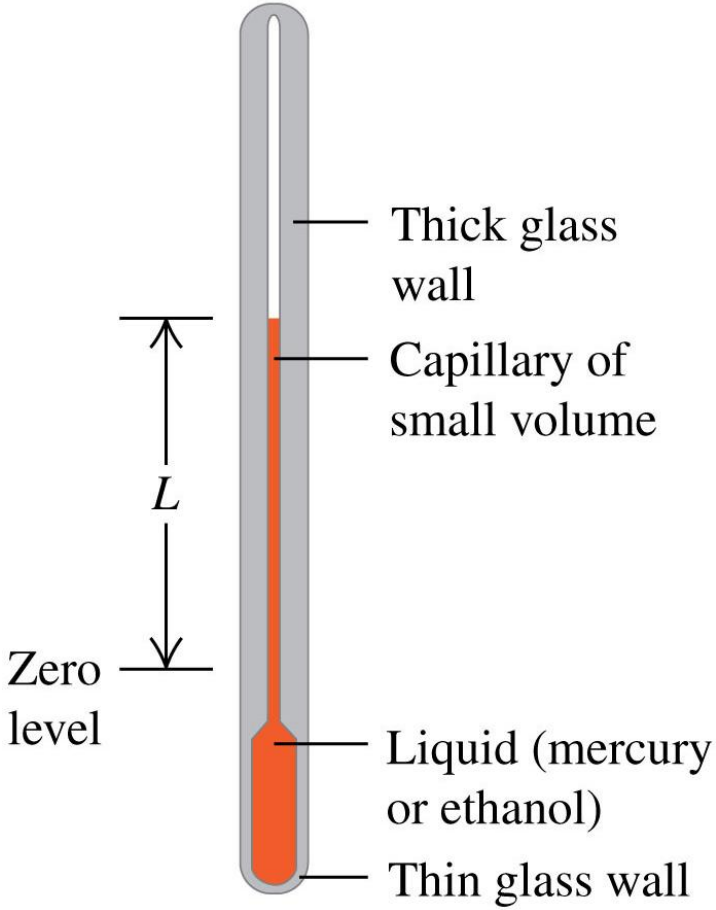
- the meaning of thermal equilibrium, and what thermometers really measure.
- the physics behind the absolute, or Kelvin, temperature scale.
- how the dimensions of an object change as a result of a temperature change.
- how to do calculations that involve heat flow, temperature changes, and changes of phase.
- how heat is transferred by conduction, convection, and radiation.

Introduction

- Does molten iron at 1500 degrees Celsius contain heat?
- The terms “temperature” and “heat” have very different meanings, even though most people use them interchangeably.
- In this chapter, we’ll focus on large-scale, or **macroscopic**, objects, but in the next chapter we’ll look at the **microscopic** scale.

Temperature and Thermal Equilibrium

- We use a **thermometer** to measure **temperature**.
- For example, the volume of the liquid in the thermometer to the right changes with temperature.
- Two systems are in **thermal equilibrium** if and only if they have the same temperature.



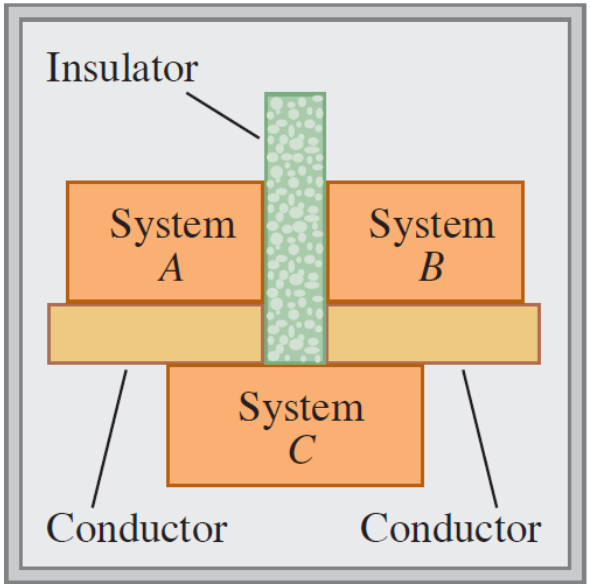
Other Types of Thermometers

- A temporal artery thermometer measures infrared radiation from the skin that overlies one of the important arteries in the head.
- Although the thermometer cover touches the skin, the infrared detector inside the cover does not.

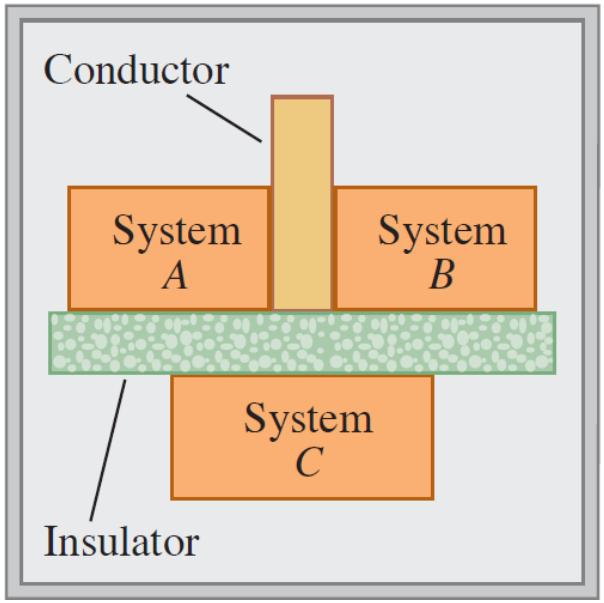
The Zeroth Law of Thermodynamics

- If *C* is initially in thermal equilibrium with both *A* and *B*, then *A* and *B* are in thermal equilibrium with each other.

(a) If systems *A* and *B* are each in thermal equilibrium with system *C* ...



(b) ... then systems *A* and *B* are in thermal equilibrium with each other.



Temperature Scales (1 of 2)

- On the **Celsius** (or **centigrade**) **temperature scale**, 0 degrees Celsius is the freezing point of pure water and 100 degrees Celsius is its boiling point.
- On the **Fahrenheit temperature scale**, 32 degrees Fahrenheit is the freezing point of pure water and 212 degrees Fahrenheit is its boiling point.
- To convert from Celsius to Fahrenheit:

$$\underbrace{T_F}_{\text{Fahrenheit temperature}} = \tfrac{9}{5}\underbrace{T_C}_{\text{Celsius temperature}} + 32^\circ$$

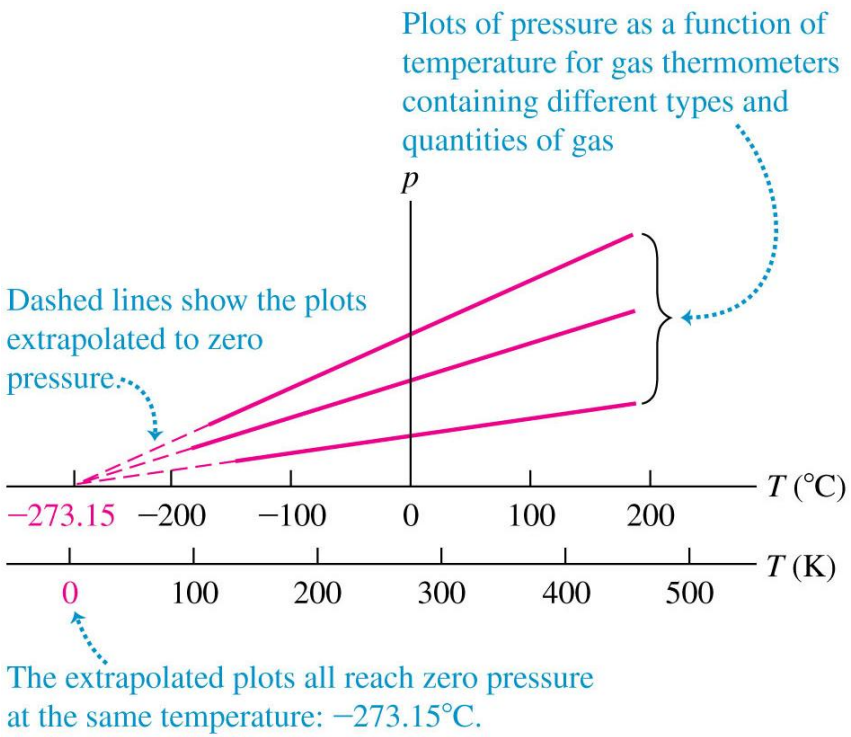
- To convert from Fahrenheit to Celsius:

$$\underbrace{T_C}_{\text{Celsius temperature}} = \tfrac{5}{9}\left(\underbrace{T_F}_{\text{Fahrenheit temperature}} - 32^\circ\right)$$

- Video Tutor Solution: Example 17.1

Absolute Zero

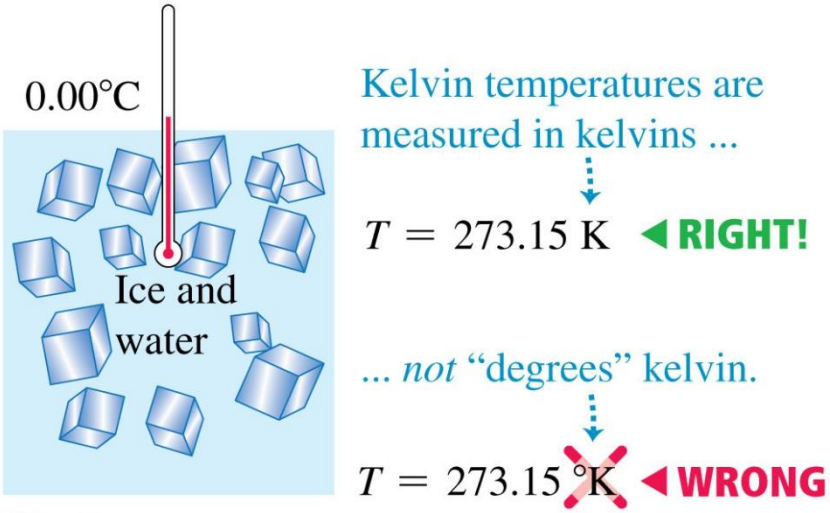
- There is a temperature, −273.15 degrees Celsius, at which the absolute pressure of any gas would become zero.



Temperature Scales (2 of 2)

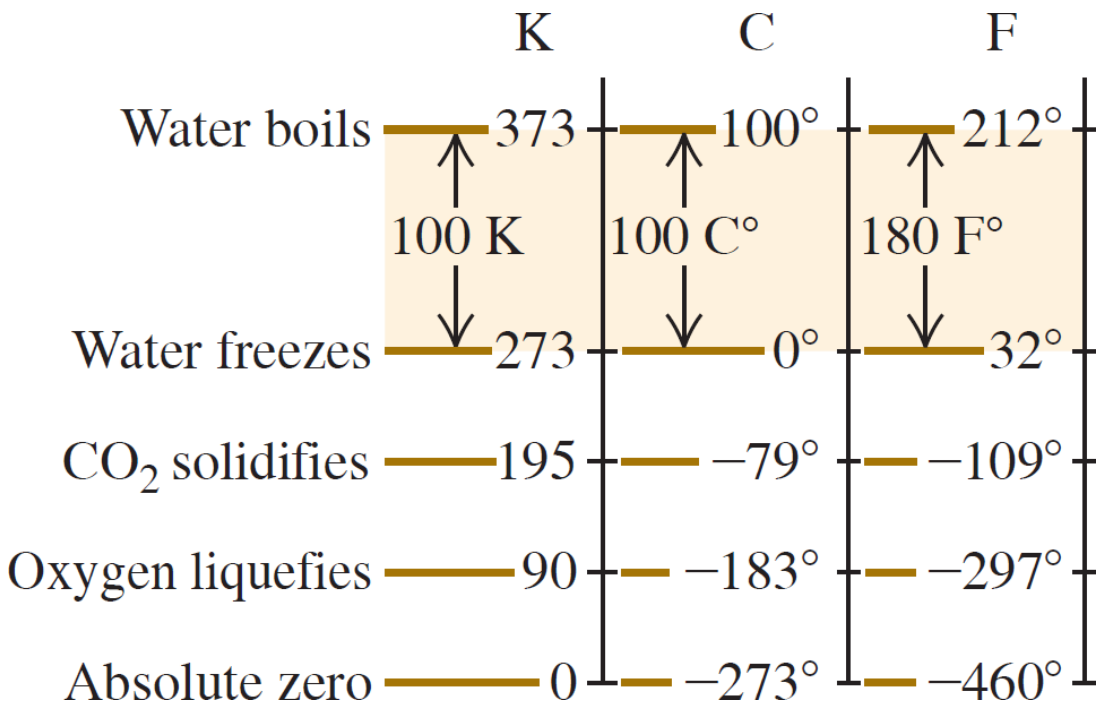
- On the **Kelvin** (or **absolute**) **temperature scale**, 0 K is the extrapolated temperature at which a gas would exert no pressure.
- To convert from Celsius to Kelvin:

Kelvin temperature → $T_K = T_C + 273.15$ ← Celsius temperature



Temperature Conversions

- Below are relationships among Kelvin (K), Celsius (C), and Fahrenheit (F) temperature scales. Temperatures have been rounded off to the nearest degree.



Question 1

0 / 10 pts

Given a temperature of 322 K, convert to degrees Fahrenheit.

Linear Thermal Expansion

- Increasing the temperature of a rod causes it to expand.
- For moderate changes in temperature, the change in length is given by:

Linear thermal expansion:

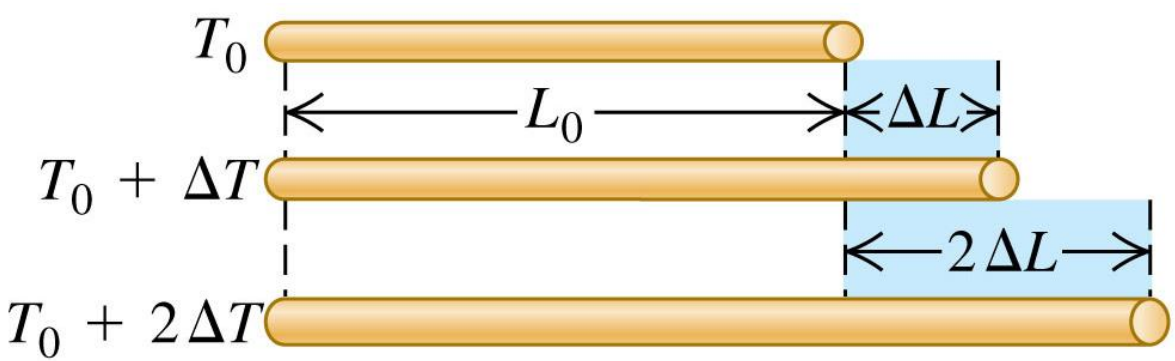
$$\Delta L = \alpha L_0 \Delta T$$

Change in length — ΔL

Original length — L_0

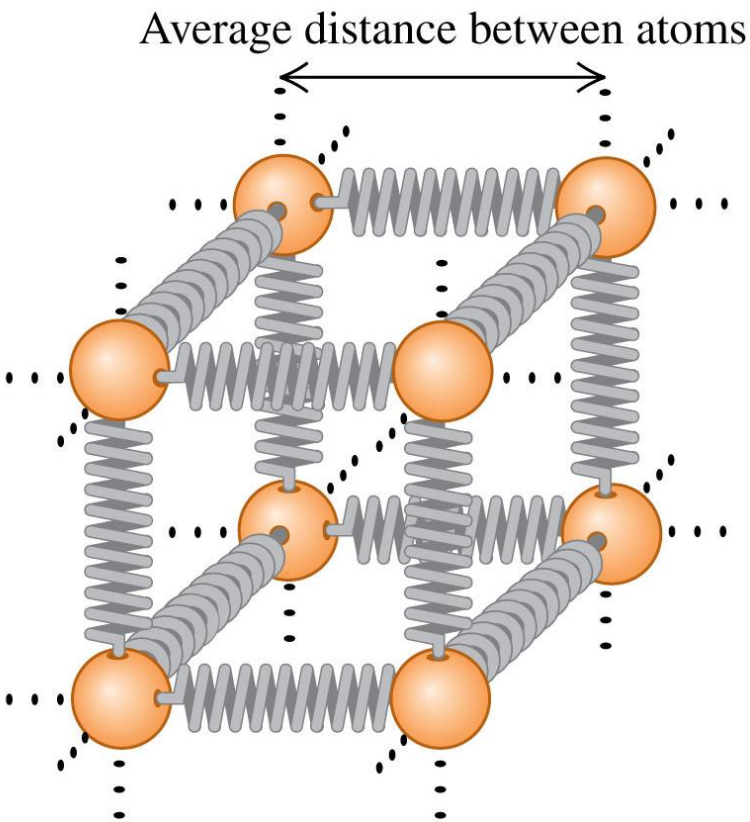
Coefficient of linear expansion — α

Temperature change — ΔT



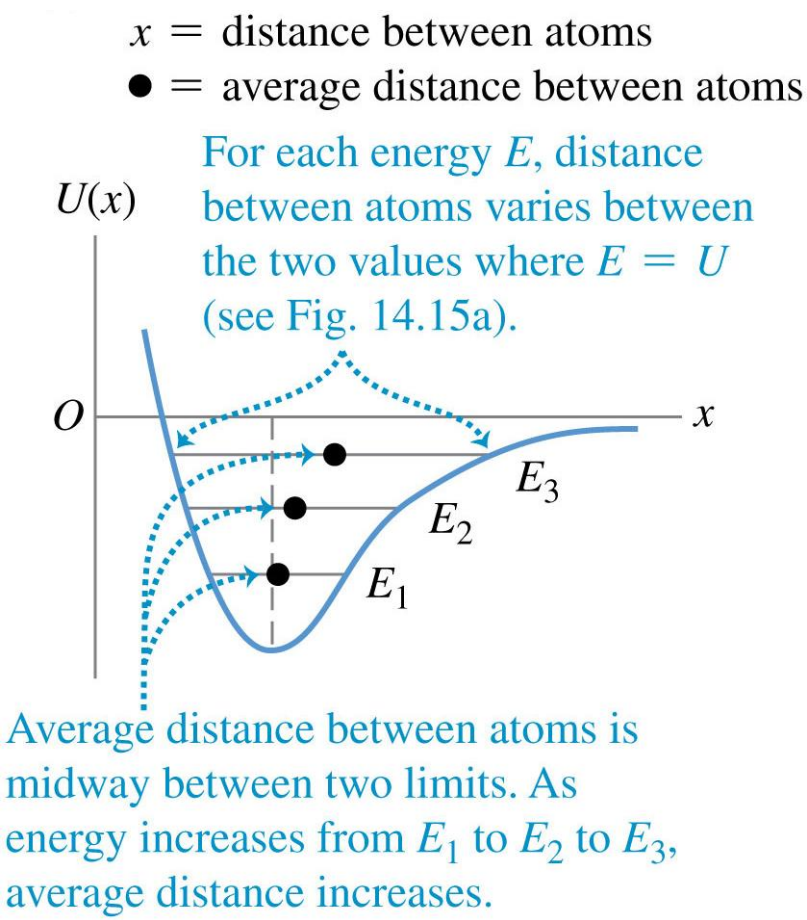
Molecular Basis for Thermal Expansion (1 of 2)

- We can understand linear expansion if we model the atoms as being held together by springs.
- When the temperature increases, the average distance between atoms also increases.
- As the atoms get farther apart, every dimension increases.



Molecular Basis for Thermal Expansion (2 of 2)

- A graph of the "spring" potential energy versus distance between neighboring atoms is not symmetrical.
- As the energy increases and the atoms oscillate with greater amplitude, the average distance increases.



Expanding Holes and Volume Expansion

- If an object has a hole in it, the hole also expands with the object, as shown.
- The hole does **not shrink**.
- The change in volume due to thermal expansion is given by

  $$\Delta V = \beta V_0 \Delta T$$

  where β is the **coefficient of volume expansion** and is equal to 3α.
- Video Tutor Solution: Example 17.3

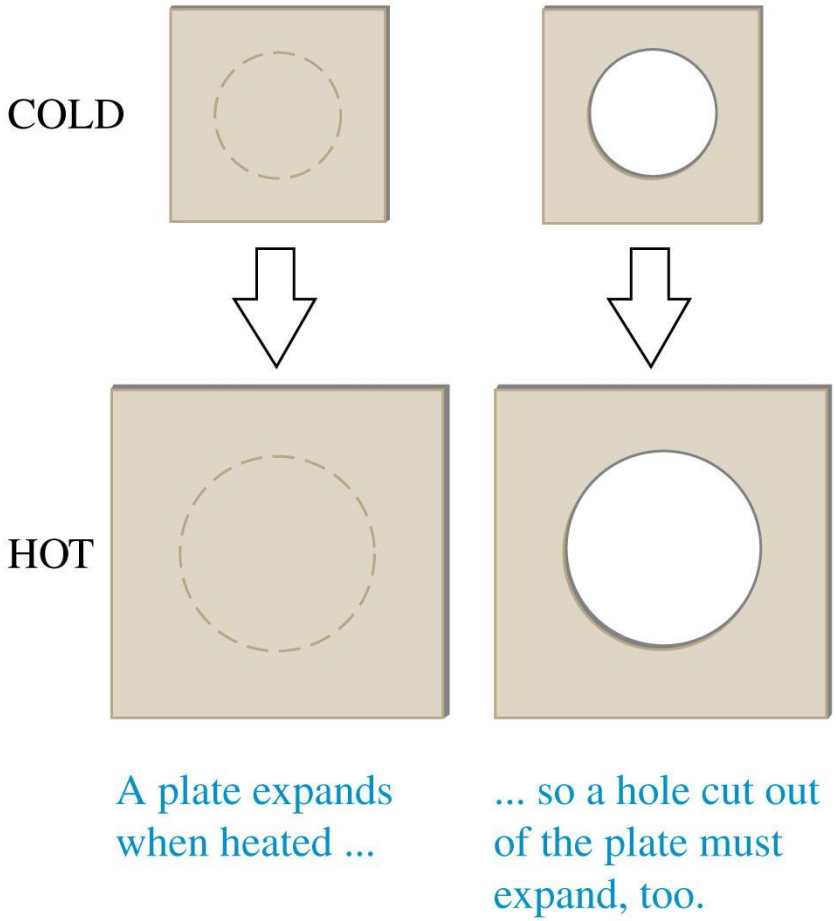


Table 17.1 Coefficients of Linear Expansion

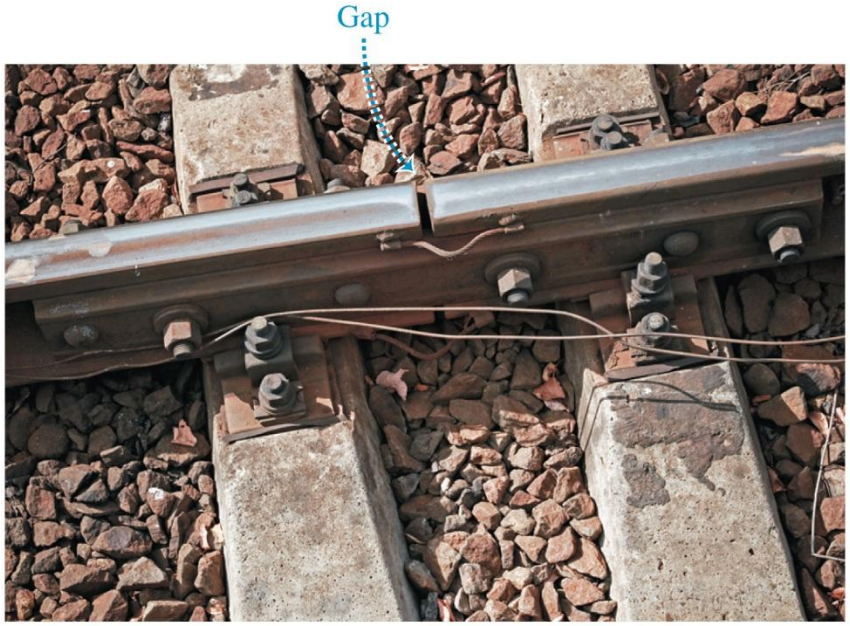
Material	$\alpha[K^{-1}$ or $(C^{\circ})^{-1}]$
Aluminum	2.4×10^{-5}
Brass	2.0×10^{-5}
Copper	1.7×10^{-5}
Glass	$0.4 - 0.9 \times 10^{-5}$
Invar (nickel–iron alloy)	0.09×10^{-5}
Quartz (fused)	0.04×10^{-5}
Steel	1.2×10^{-5}

Table 17.2 Coefficients of Volume Expansion

Solids	$\beta[K^{-1}$ or $(C^{\circ})^{-1}]$
Aluminum	7.2×10^{-5}
Brass	6.0×10^{-5}
Copper	5.1×10^{-5}
Glass	$1.2 - 2.7 \times 10^{-5}$
Invar	0.27×10^{-5}
Quartz (fused)	0.12×10^{-5}
Steel	3.6×10^{-5}

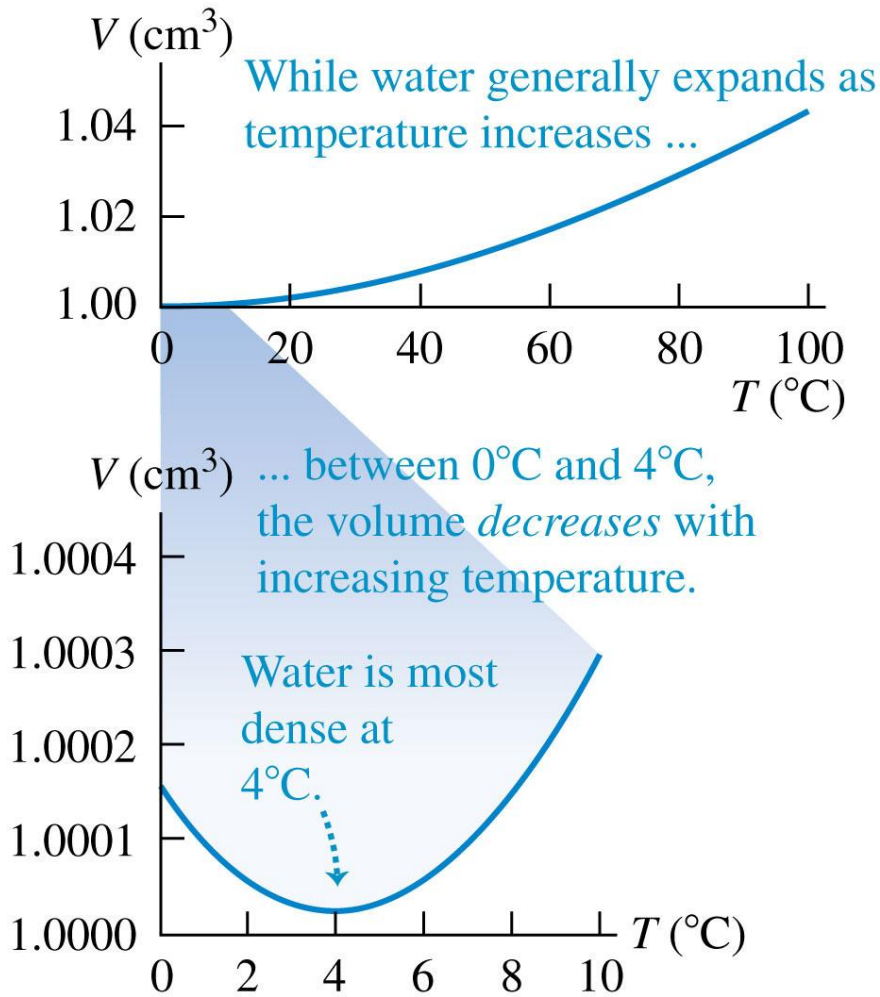
Example of Thermal Expansion

- This railroad track has a gap between segments to allow for thermal expansion.
- On hot days, the segments expand and fill in the gap.
- If there were no gaps, the track could buckle under very hot conditions.



Thermal Expansion of Water

- Between 0 degrees Celsius and 4 degrees Celsius, water **decreases** in volume with increasing temperature.
- Because of this anomalous behavior, lakes freeze from the top down instead of from the bottom up.



Question 2

0 / 10 pts

A substance is cast into a bar 10.9 m long. When the bar's temperature is raised from 9°C to 56°C, its length increases by 0.025 m. Calculate the coefficient of linear expansion for this material and supply the missing coefficient in the blank below.

α = _____ $\times 10^{-5}$ $(C°)^{-1}$

Question 3

0 / 10 pts

A metal has a coefficient of linear expansion of 2.5×10^{-5} $(C°)^{-1}$. Suppose a pipe made of this metal is 1.9 m long. Calculate the temperature increase in C° if its length increases by 3.3 mm.

Question 4

0 / 10 pts

Suppose a glass beaker with a linear expansion coefficient of 1.6×10^{-5} $(C°)^{-1}$ is filled to the brim with 110 cm^3 of a liquid with a volume expansion coefficient of 3.9×10^{-4} $(C°)^{-1}$. If the beaker is warmed by 23 C°, how much fluid in mm^3 spills on the lab table? (Be careful with units!)

V = _____ mm^3

Thermal Stress

- If we change the temperature of a rod but prevent it from expanding or contracting, **thermal stress** develops.

Thermal stress:

$$\frac{F}{A} = -Y\alpha\,\Delta T$$

Force needed to keep length of rod constant

Young's modulus

Temperature change

Coefficient of linear expansion

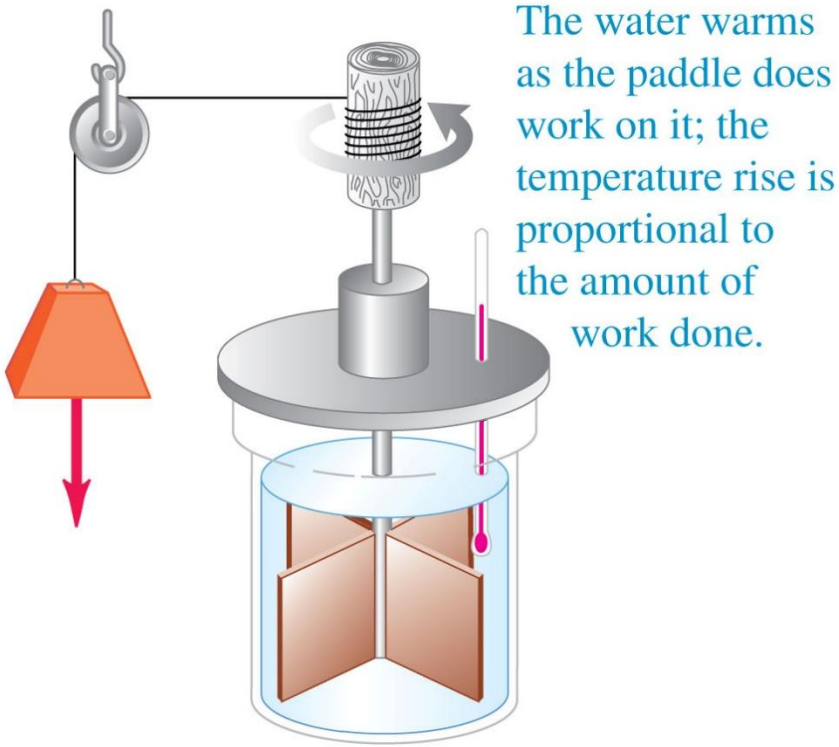
Cross-sectional area of rod

- Expansion joints on bridges are needed to accommodate changes in length that result from thermal expansion.

- Video Tutor Solution: Example 17.4

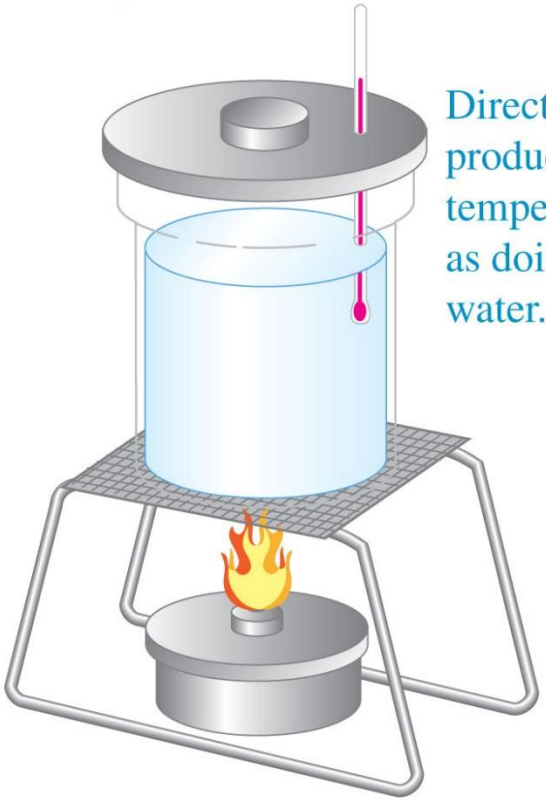
Quantity of Heat (1 of 2)

- Sir James Joule (1818–1889) studied how water can be warmed by vigorous stirring with a paddle wheel.



Quantity of Heat (2 of 2)

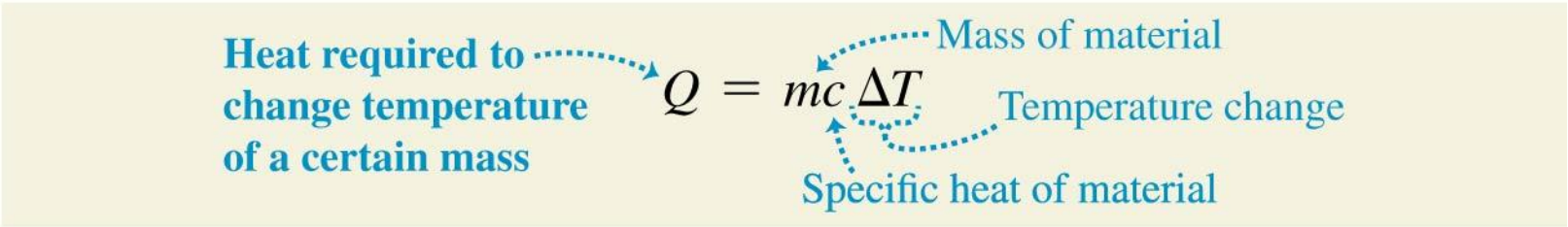
- The same temperature change caused by stirring can also be caused by putting the water in contact with some hotter object.
- The **calorie** (abbreviated cal) is the amount of heat required to raise the temperature of 1 gram of water from 14.5 degrees Celsius to 15.5 degrees Celsius.



Direct heating can produce the same temperature change as doing work on the water.

Specific Heat

- The quantity of heat Q required to increase the temperature of a mass m of a certain material by ΔT is:



- The **specific heat** c has different values for different materials.
- The specific heat of water is approximately $4190\ \mathrm{J/kg \cdot K}$.

Molar Heat Capacity

- The quantity of heat Q required to increase the temperature of n moles of a certain material by ΔT is:

Heat required to change temperature of a certain number of moles → $Q = nC\Delta T$

Number of moles of material

Temperature change

Molar heat capacity of material

- The **molar heat capacity** C has different values for different materials.
- The molar heat capacity of water is approximately 75.4 J/mol·K.
- Video Tutor Demonstration: Heating Water and Aluminum

Table 17.3 Specific Heats and Molar Heat Capacities

Substance	Specific Heat, *c* $(J/kg \cdot K)$	Molar Mass, *M* (kg/mol)	Molar Heat Capacity, *C* $(J/mol \cdot K)$
Aluminum	910	0.0270	24.6
Beryllium	1970	0.00901	17.7
Copper	390	0.0635	24.8
Ethanol	2428	0.0461	111.9
Ethylene glycol	2386	0.0620	148.0
Ice (near 0°C)	2100	0.0180	37.8
Iron	470	0.0559	26.3
Lead	130	0.207	26.9
Marble (CaCO3)	879	0.100	87.9
Mercury	138	0.201	27.7
Salt (NaCl)	879	0.0585	51.4
Silver	234	0.108	25.3
Water (liquid)	4190	0.0180	75.4

Question 5

0 / 10 pts

A bottle filled with 7.2-L of water falls from a height of 245.1 m and hits the ground ($g = 9.8$ m/s^2). If all the mechanical energy loss of the water went into heating the water, find its **change** in temperature (in C^o).

Question 6

0 / 10 pts

A given substance has a mass of 2 kg and specific heat of 434 J/(kg·C°). How much energy in kJ is required to raise the temperature of the substance by 17 C°?

Q = _____kJ

Question 7

0 / 10 pts

Suppose 2,017 J of energy are transferred to a block of mass 1.6 kg and specific heat 447 J/(kg·C°). Calculate the temperature change in Celsius degrees.

Question 8

0 / 10 pts

Suppose 4.5 kg of chloroform at 21.8°C is poured into 10.5 kg of propylene glycol at 40.4°C. Calculate the final equilibrium temperature, neglecting any energy processes associated with mixing, given c_{prop} = 2500 J/(kg·K) and $c_{chloroform}$ = 1050 J/(kg·K).

T = _____ °C

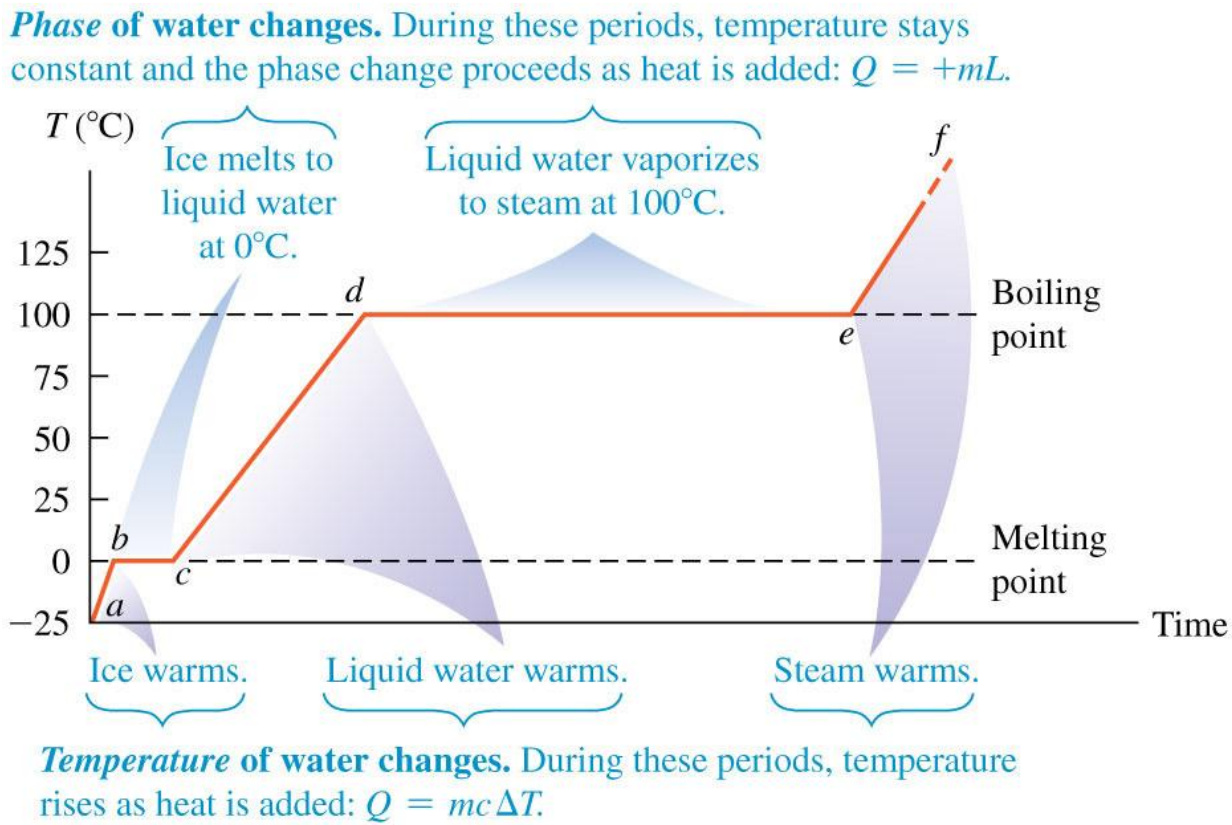
Phase Changes

- The **phases** (or states) of matter are solid, liquid, and gas.
- A **phase change** is a transition from one phase to another.
- The temperature does not change during a phase change.
- The **latent heat**, L, is the heat per unit mass that is transferred in a phase change.

Heat transfer in a phase change $Q = \pm mL$

- m: Mass of material that changes phase
- L: Latent heat for this phase change
- $\pm$: + if heat enters material, − if heat leaves

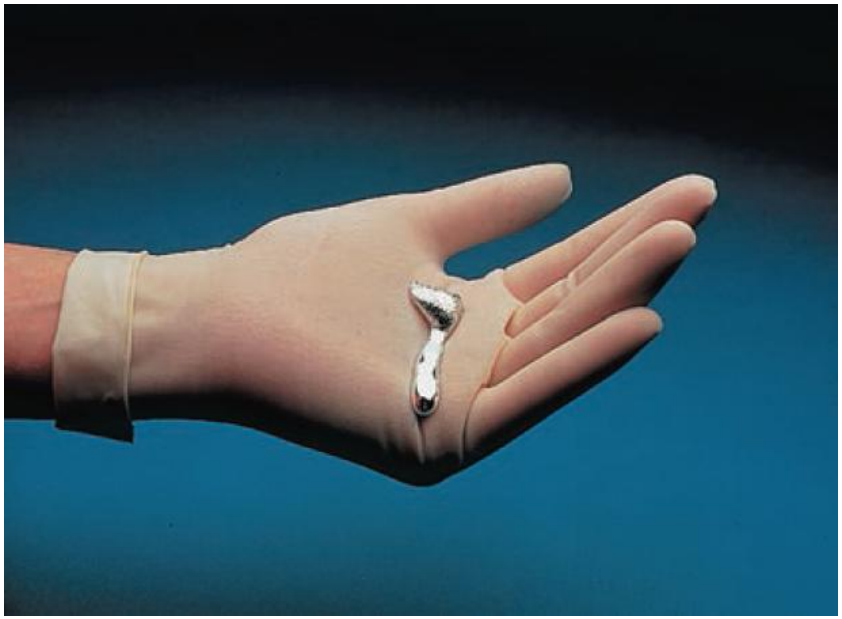
Heat Added to Ice at a Constant Rate



- Video Tutor Solution: Example 17.8

Heat of Fusion

- The metal gallium, shown here melting in a person's hand, is one of the few elements that melts at room temperature.
- Its melting temperature is 29.8 degrees Celsius, and its **heat of fusion** is $L_f = 8.04 \times 10^4$ J/kg.



Heat of Vaporization

- The water may be warm and it may be a hot day, but these children will feel cold when they first step out of the swimming pool.
- That's because as water evaporates from their skin, it removes the **heat of vaporization** from their bodies.
- To stay warm, they will need to dry off immediately.

Question 9

0 / 10 pts

How much energy (in kJ) is required to convert 1.92 kg of ice at -12°C to water at 24°C?

[Use: c_{ice} = 2100 J/(kg·K); c_{water} = 4190 J/(kg·K); $L_{f,water}$ = 334x10^3 J/kg]

Question 10

0 / 10 pts

How much energy in MJ is required to turn 5.5 kg of ice at 0°C to steam at 100°C?

[Use: L_f = 335,000 J/kg; L_v = 2,260,000 J/kg; c = 4190 J/(kg·K).]

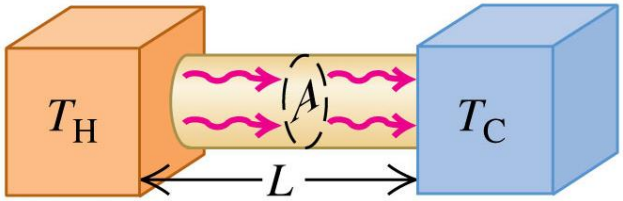
Q = _____MJ

Mechanisms of Heat Transfer

- In nature, energy naturally flows from higher temperature objects to lower temperature objects; this is called **heat transfer**.
- The three mechanisms of heat transfer are **conduction**, **convection**, and **radiation**.
- **Conduction** occurs within an object or between two objects in contact.
- **Convection** depends on motion of mass from one region of space to another.
- **Radiation** is heat transfer by electromagnetic radiation, such as sunshine, with no need for matter to be present in the space between objects.

Conduction of Heat

- In conduction, heat flows from a higher to a lower temperature.
- Consider a solid rod of conducting material with cross-sectional area A and length L.
- The left end of the rod is kept at a temperature T_H and the right end at a lower temperature T_C.



- The rate that heat is transferred is:

Rate of heat flow

Temperatures of hot and cold ends of rod

Heat current in conduction

$$H = \frac{dQ}{dt} = kA\frac{T_H - T_C}{L}$$

Length of rod

Thermal conductivity of rod material

Cross-sectional area of rod

Thermal Conductivities of Some Common Substances

Substance	$k(W/m \cdot K)$
Silver	406
Copper	385
Aluminum	205
Wood	0.12 – 0.04
Concrete	0.8
Fiberglass	0.04
Styrofoam	0.027

Question 12

0 / 10 pts

The wall of a room is composed of 10 cm of brick with thermal conductivity 0.3 W/(m·K). Drywall on the inside surface of the brick has a thickness of 2 cm and thermal conductivity [0.01 W/(m·K). If the temperature outside is -5°C and the temperature inside is 21°C, calculate the magnitude of the rate at which energy is going through the wall, from inside to outside, if the area of the wall is 10 m^2.

[Hint: The heat current in conduction through a single layer is $H = \frac{k}{L} A \left(T_H - T_C\right) = \frac{A}{R}\left(T_H - T_C\right)$. For multiple stacked layers, the effective R value is the sum of individual R values.]

Question 13

0 / 10 pts

A ball hanging by a wire has a radius of 1.8 m and emissivity 0.67. If the surface temperature of the ball is 105°C and the room is at temperature 26°C, calculate the net rate in kW at which energy is radiated by the ball.

H = _____kW

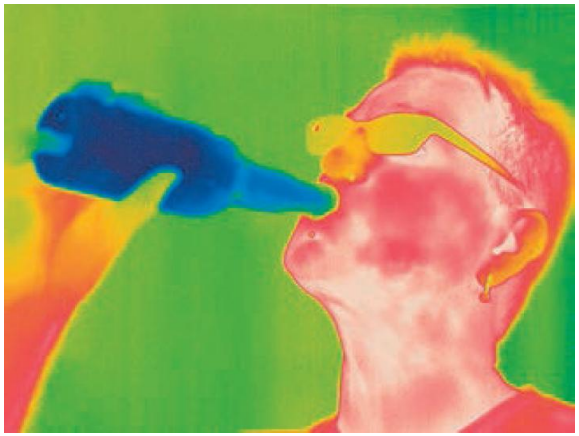
Convection of Heat

- **Convection** is the transfer of heat by the mass motion of fluid.
- A heating element in the tip of this submerged tube warms the surrounding water, producing a complex pattern of free convection.

- Video Tutor Demonstration: Candle Chimneys

Radiation of Heat

- **Radiation** is the transfer of heat by electromagnetic waves, such as visible light or infrared.
- This false-color infrared photograph reveals radiation emitted by various parts of the man's body.
- The strongest emission comes from the warmest areas, while there is very little emission from the bottle of cold beverage.
- **Stefan-Boltzmann law**: The **heat current** in radiation is:



$$H = Ae\sigma T^4$$

Heat current in radiation → H

Area of emitting surface → A

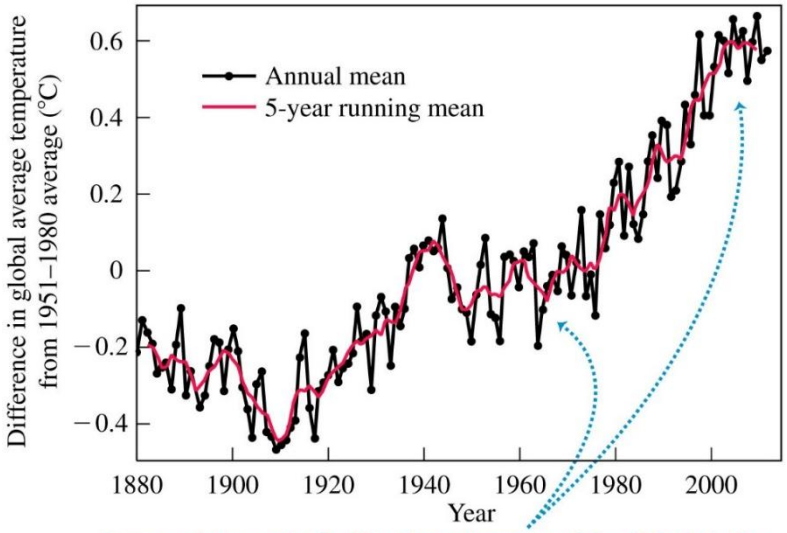
Emissivity of surface → e

Stefan–Boltzmann constant → σ

Absolute temperature of surface → T

Radiation and Climate Change

- The energy radiated by the earth's surface is mostly infrared.
- CO_2 molecules in our atmosphere readily absorb some of this infrared radiation and reradiate part of it back down toward the surface.



Increased atmospheric CO_2 due to burning of fossil fuels is the cause of this continuing increase in global average temperatures.